733. Flood Fill

Solved Easy

You are given an image represented by an m x n grid of integers image, where image[i][j] represents the pixel value of the image. You are also given three integers sr, sc, and color. Your task is to perform a **flood fill** on the image starting from the pixel image[sr][sc].

To perform a **flood fill**:

1. Begin with the starting pixel and change its color to color.
2. Perform the same process for each pixel that is **directly adjacent** (pixels that share a side with the original pixel, either horizontally or vertically) and shares the **same color** as the starting pixel.
3. Keep **repeating** this process by checking neighboring pixels of the *updated* pixels and modifying their color if it matches the original color of the starting pixel.
4. The process **stops** when there are **no more** adjacent pixels of the original color to update.

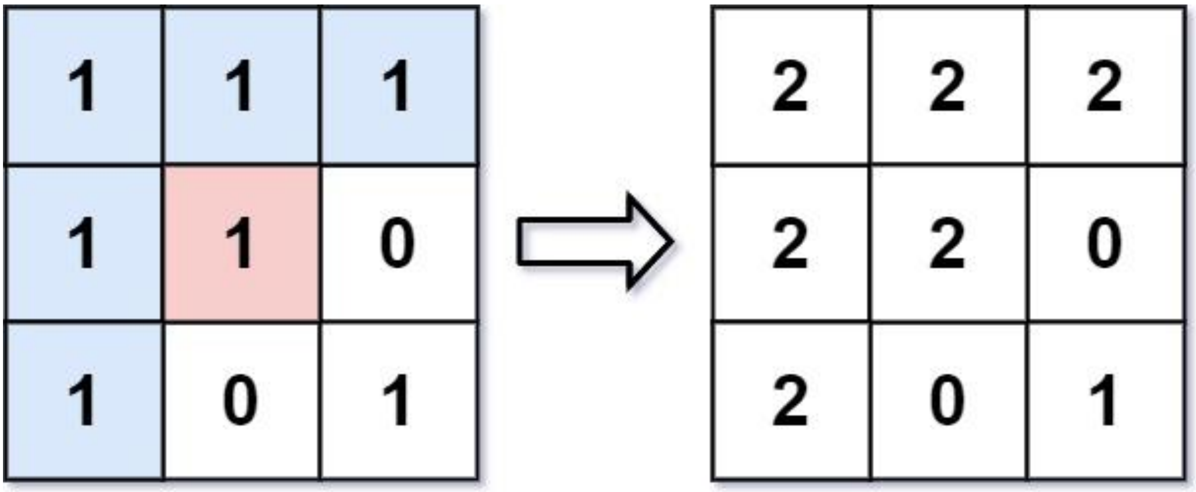
Return the **modified** image after performing the flood fill.

Example 1:

Input: image = [[1,1,1],[1,1,0],[1,0,1]], sr = 1, sc = 1, color = 2

Output: [[2,2,2],[2,2,0],[2,0,1]]

Explanation:



From the center of the image with position (sr, sc) = (1, 1) (i.e., the red pixel), all pixels connected by a path of the same color as the starting pixel (i.e., the blue pixels) are colored with the new color.

Note the bottom corner is **not** colored 2, because it is not horizontally or vertically connected to the starting pixel.

Example 2:

Input: image = [[0,0,0],[0,0,0]], sr = 0, sc = 0, color = 0

Output: [[0,0,0],[0,0,0]]

Explanation:

The starting pixel is already colored with 0, which is the same as the target color. Therefore, no changes are made to the image.

Constraints:

- m == image.length
- n == image[i].length
- 1 <= m, n <= 50
- 0 <= image[i][j], color < 2^{16}
- 0 <= sr < m
- 0 <= sc < n